

Can AI Make a “Thinking Partner” for Simulated Young Adult Agents

How do LLMs shape opinion formation when used as "thinking partners" by simulated young adult AI agents?

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INTRODUCTION

Large Language Models (LLMs) are increasingly being used as “thinking partners”, a conversational AI system that actively participates in users’ reasoning. With 64% of **young adults** (ages 16–24) already using generative AI regularly, a key concern is whether different AI “thinking partner” styles help or hinder responsible opinion formation. Three styles are investigated: **Steelman** (adversarial argumentation), **Socratic** (reasoning-based questioning), and **Neutral** (balanced information)

HYPOTHESES

- H1:** The Steelman condition will produce significantly greater opinion change than the Socratic or Neutral conditions
- H2:** The Neutral condition will produce higher epistemic trust than the Steelman or Socratic conditions
- H3:** The Socratic condition will produce higher perceived autonomy than the Steelman or Neutral conditions

METHODOLOGY

- Between-subject study
- 159 simulated young adult personas
- 3 Independent Variables: Steelman, Socratic, Neutral
- 3 Dependent Variables: Opinion Change, Epistemic Trust, Epistemic Autonomy
- Conversation Topic: Individual vs Systemic Action for Climate Change
- Thinking Partners: LLaMA 3.1:b
- Personas: LLaMA 3.1 (8.03B parameters)



ANALYSIS

Quantitative

- One-way ANOVA per outcome variable
- Kruskal-Wallis robustness check
- Tukey HSD post-hoc for significant results

Qualitative

Thematic Analysis:

Opinion Change	Epistemic Trust	Epistemic Autonomy
T1: Systematic Shift T2: Individual Shift T3: Together	TR1: No Doubt TR2: Factual Doubt TR3: Framing Doubt	A1: Clear Augmentation A2: Dependent Augmentation

RESULTS

ANOVA Results

Variable	F(2, 152)	p-value	η^2
Opinion Change (SQ1)	35.46	< .001	.318
Epistemic Trust (SQ2)	2.02	.136	.026
Epistemic Autonomy (SQ3)	2.78	.065	.035

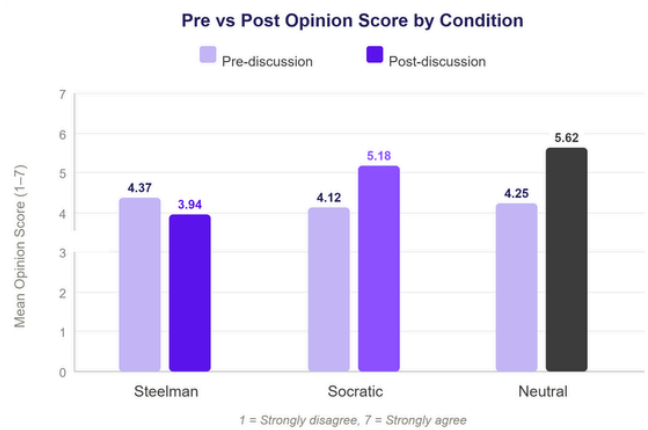
Thematic Analysis Frequencies

Theme	Steelman	Socratic	Neutral
OPINION CHANGE			
T1 Systematic Shift	31/43	15/39	21/39
T2 Individual Shift	6/43	17/39	15/39
T3 Together	4/43	3/39	11/39
EPISTEMIC TRUST			
TR1 No Doubt	27/36	23/24	32/34
TR2/TR3 Doubt Present	9/36	1/24	1/34
EPISTEMIC AUTONOMY			
A1 Clear Augmentation	38/41	35/38	37/39

DISCUSSIONS

The only statistically significant finding was the **Opinion Change** variable ($F(2,152) = 35.46$, $p < .001$, $\eta^2 = .318$). The Steelman condition produced a negative mean shift ($M=-0.42$), moving personas to systematic change, while Socratic ($M=+1.06$) and Neutral ($M=+1.37$) produced comparable positive shifts. The qualitative findings support this claim as well. The **T1** theme dominated Steelman responses, indicating that strong argumentation pushed personas to systemic change over individual action. Both **Epistemic Trust** ($F(2,152) = 2.02$, $p = .136$) and **Epistemic Autonomy** showed no significant difference across all three conditions, with themes **TR1** and **A1** dominating in all conditions.

Post-hoc comparison confirmed that Steelman differed significantly from Socratic (diff = -1.48, $p < .001$) and Neutral (diff = -1.79, $p < .001$), while Socratic and Neutral did not differ from each other (diff = -0.31, $p = .373$).



H	Description	Verdict
H1	Steelman produces greatest opinion change	Partially Supported
H2	Neutral produces highest epistemic trust	Not Supported
H3	Socratic produces highest epistemic autonomy	Not Supported

LIMITATIONS & FUTURE WORK

- Simulated personas lack human variability – results are only exploratory
- Single session format only captures immediate effect – no longitudinal data
- 121/477 potential qualitative responses are missing; the remaining answers showed notable uniformity

Future Work should replicate the experiment with human participants, extend to multiple sessions with multiple topics